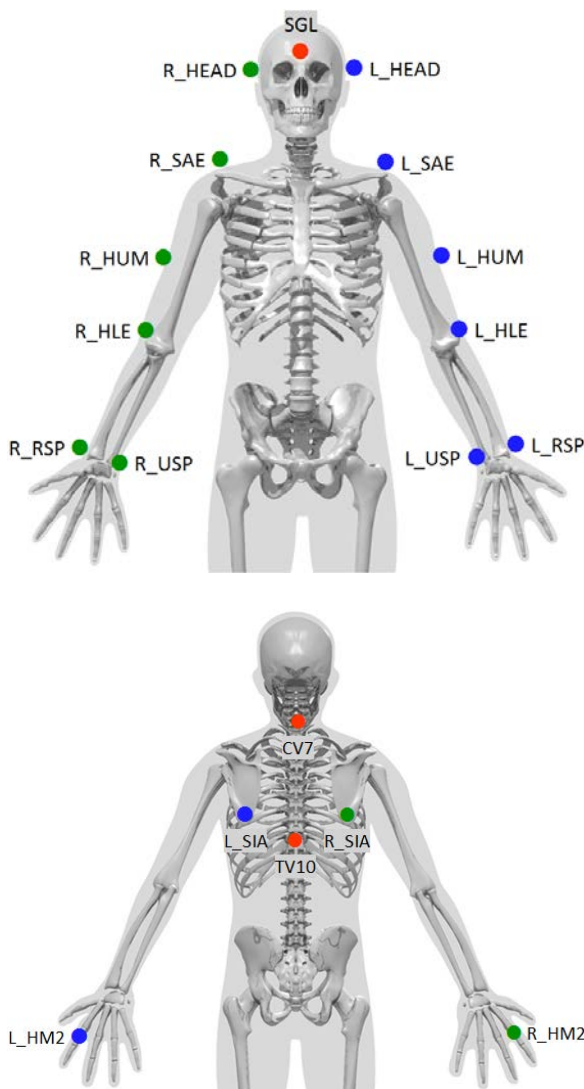


Qualisys PAF package: Upper body CAST marker set



Name	Ref. ¹	Location	Static (17)	Dyn. (17)
L_HEAD		Just above the ear	X	X
R_HEAD		Just above the ear	X	X
SGL	SGL	Glabulla	X	X
CV7	CV7	Cervical Vertebrae	X	X
L_SIA	SIA	Scapula - Inferior Angle	X	X
R_SIA	SIA	Scapula - Inferior Angle	X	X
TV10	TV10	Thoracic Vertebrae	X	X
L_SAE	SAE	Scapula - Acromial Edge	X	X
L_HUM			X	X
L_HLE	HLE	Humerus - Lateral Epicondyle	X	X
L_RSP	RSP	Radius - Styloid Process	X	X
L_USP	UPS	Ulna - Styloid Process	X	X
L_HM2	HM2	Basis of Forefinger	X	X
R_SAE	SAE	Scapula - Acromial Edge	X	X
R_HUM			X	X
R_HLE	HLE	Humerus - Lateral Epicondyle	X	X
R_RSP	RSP	Radius - Styloid Process	X	X
R_USP	USP	Ulna - Styloid Process	X	X
R_HM2	HM2	Basis of Forefinger	X	X

¹ Sint Jan, S. Van (2007). Color Atlas of Skeletal Landmark Definitions. Guidelines for Reproducible Manual and Virtual Palpations. Edinburgh: Churchill Livingstone.

Qualisys PAF package: CAST lower body marker set¹

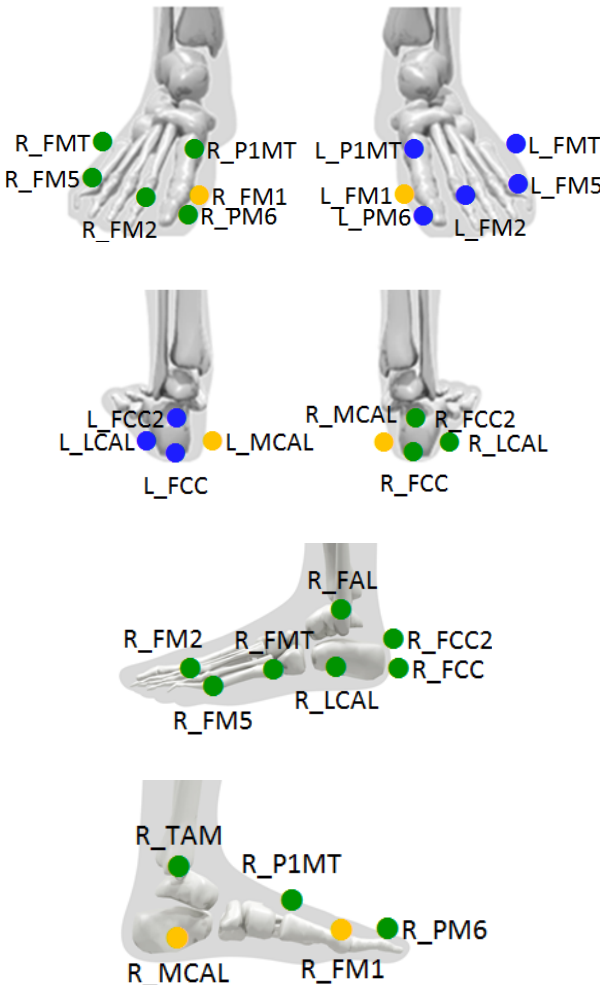
The image displays two anatomical diagrams of a human lower body, showing the placement of 36 markers. The top diagram is a posterior view (back) of the lower body, and the bottom diagram is an anterior view (front). Markers are represented by colored dots (green, blue, yellow) and labeled with codes. The back view includes markers for the pelvis (R_IAS, L_IAS), thighs (R_TH1-4, L_TH1-4), knees (R_FLE, L_FLE, R_FME, L_FME), and ankles (R_SK1-4, L_SK1-4, R_FAL, L_FAL, R_TAM, L_TAM). The front view includes markers for the pelvis (L_IPS, R_IPS), ankles (L_FCC, R_FCC), and feet (R_FM1-5, L_FM1-5).

Name	Ref. ²	Location	Static (36)	Dyn. (28)
L_IAS	IAS	Anterior superior iliac spine	X	X
L_IPS	IPS	Posterior superior iliac spine	X	X
R_IPS	IPS	Posterior superior iliac spine	X	X
R_IAS	IAS	Right anterior superior iliac spine	X	X
L_TH1-4		Cluster	X	X
L_FLE	FLE	Lateral epicondyle	X	
L_FME	FME	Medial epicondyle	X	
L_SK1-4		Cluster	X	X
L_FAL	FAL	Lateral prominence of the lateral malleolus	X	
L_TAM	TAM	Medial prominence of the medial malleolus	X	
L_FCC	FCC	Aspect of the Achilles tendon insertion on the calcaneus	X	X
L_FM1	FM1	Dorsal margin of the first metatarsal head	X	X
L_FM2	FM2	Dorsal aspect of the second metatarsal head	X	X
L_FM5	FM5	Dorsal margin of the fifth metatarsal head	X	X
R_TH1-4		Cluster	X	X
R_FLE	FLE	Lateral epicondyle	X	
R_FME	FME	Medial epicondyle	X	
R_SK1-4		Cluster	X	X
R_FAL	FAL	Lateral prominence of the lateral malleolus	X	
R_TAM	TAM	Medial prominence of the medial malleolus	X	
R_FCC	FCC	Aspect of the Achilles tendon insertion on the calcaneus	X	X
R_FM1	FM1	Dorsal margin of the first metatarsal head	X	X
R_FM2	FM2	Dorsal aspect of the second metatarsal head	X	X
R_FM5	FM5	Dorsal margin of the fifth metatarsal head	X	X

¹ Cappozzo, A., Catani, F., Della Croce, U., Leardini, A. (1995). Position and orientation in space of bones during movement: anatomical frame definition and determination. Clinical Biomechanics, 10, 171-178.

² Sint Jan, S. Van (2007). Color Atlas of Skeletal Landmark Definitions. Guidelines for Reproducible Manual and Virtual Palpations. Edinburgh: Churchill Livingstone.

Qualisys PAF package: CAST¹ lower body marker set (including Oxford foot model³)

	Name	Ref. ²	Location	Static (20)	Dyn. (16)
	L_FCC2		Proximal end of posterior aspect of calcaneus	X	X
	L_FCC	FCC	Distal end of posterior aspect of calcaneus	X	X
	L_LCAL		Lateral calcaneus (placed on the lateral aspect of the calcaneus, at the same height and distance as L_MCAL) ³	X	X
	L_FMT	FMT	Tuberosity of the fifth metatarsal bone	X	X
	L_FM5	FM5	Base of fifth metatarsal (along the line through the distal heads of the first and fifth metatarsal heads) ³	X	X
	L_FM2	FM2	Dorsal margin of the second metatarsal head	X	X
	L_PM6	PM6	Proximal medial phalanx of the big toe	X	X
	L_FM1	FM1	Base of first metatarsal (along the line through the distal heads of the first and fifth metatarsal heads) ³	X	
	L_P1MT		Base of the first metatarsal bone	X	X
	L_MCAL		Medial calcaneus (placed on the medial aspect of the calcaneus, at the same height and distance as L_LCAL) ³	X	
	R_FCC2		Proximal end of posterior aspect of calcaneus	X	X
	R_FCC	FCC	Distal end of posterior aspect of calcaneus	X	X
	R_LCAL		Lateral calcaneus (placed on the lateral aspect of the calcaneus, at the same distance as R_MCAL) ³	X	X
	R_FMT	FMT	Tuberosity of the fifth metatarsal bone	X	X
	R_FM5	FM5	Base of fifth metatarsal (along the line through the distal heads of the first and fifth metatarsal heads) ³	X	X
	R_FM2	FM2	Dorsal margin of the second metatarsal head	X	X
	R_PM6	PM6	Proximal medial phalanx of the big toe	X	X
	R_FM1	FM1	Base of first metatarsal (along the line through the distal heads of the first and fifth metatarsal heads) ³	X	
	R_P1MT		Base of the first metatarsal bone	X	X
	R_MCAL		Medial calcaneus (placed on the medial aspect of the calcaneus, at the same height and distance as R_LCAL) ³	X	

¹ Cappozzo, A., Catani, F., Della Croce, U., Leardini, A. (1995). Position and orientation in space of bones during movement: anatomical frame definition and determination. Clinical Biomechanics, 10, 171-178.

² Sint Jan, S. Van (2007). Color Atlas of Skeletal Landmark Definitions. Guidelines for Reproducible Manual and Virtual Palpations. Edinburgh: Churchill Livingstone.

³ Stebbins J, Harrington M, Thompson N, Zavatsky A, Theologis T. (2006) Repeatability of a model for measuring multi-segment foot kinematics in children. Gait Posture. 2006 Jun;23(4):401-10.

Qualisys PAF module: Assembly instructions for marker clusters (Art. 160145)



Top screw hole

Bottom screw hole

The marker clusters must be assembled before their first use.

The clusters have four screw wholes at the top end, but two screw holes at their bottom end.

We recommend assembling them as described below, as this gives each leg cluster a unique shape.



Name	Note
Pelvis 1	Top screw hole
Pelvis 2	Top screw hole
Pelvis 3	
Pelvis 4	
R_THI 1	Top screw hole
R_THI 2	Bottom screw hole
R_THI 3	
R_THI 4	
R_SHANK 1	Bottom screw hole
R_SHANK 2	Top screw hole
R_SHANK 3	
R_SHANK 4	
L_THI 1	Top screw hole
L_THI 2	Top screw hole
L_THI 3	
L_THI 4	
L_SHANK 1	Bottom screw hole
L_SHANK 2	Bottom screw hole
L_SHANK 3	
L_SHANK 4	